

HITIK PANCHAL

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SUMMARY

Computer Science Master's student at Universität Bonn (Grade: 1.7) with 1.5 years of production experience building high-throughput backend systems at NSE India. Hands-on professional experience with Python, Java/Spring Boot, SQL, and REST APIs across large-scale financial systems. University research focused on NLP and LLMs - including LoRA fine-tuning of transformer models and multilingual narrative classification using RoBERTa and LLaMA. Personal projects spanning full-stack websites, agile planners and tableau dashboards.

PROFESSIONAL EXPERIENCE

Associate System Analyst · NSEIT Private Limited

May 2023 – Feb 2024 · Mumbai, IN

- **Project lead** for NCMS CM module - owned delivery planning, stakeholder communication, and release coordination across a cross-functional team of engineers and QA analysts.
- Architected and delivered migration of NSE Clearing Management System (NCMS) from Java/Spring Boot monolith to REST API-based architecture, increasing throughput by ~20% and enabling the platform to sustain **20,000+ trades/minute** for live clearing operations.
- Identified and resolved critical backend performance bottlenecks; DB query refactoring reduced database load by ~10% and eliminated several production bugs, improving overall system stability by **4-5%**.
- Mentored 2 junior developers through structured code reviews, improving code quality standards and reducing review cycle time across the NCMS module.

Stack: Java, Spring Boot, MySQL, JavaScript, Linux, REST APIs, Postman, Git

Trainee Associate System Analyst · NSEIT Private Limited

Nov 2022 – May 2023 · Mumbai, IN

- Executed production deployments on Linux for NCMS serving **100,000+ end users**, maintaining near-zero downtime across all release cycles.
- **Led** Git migration of 10+ legacy NCMS codebases, standardising version control across the team and significantly reducing branch conflicts and manual merge overhead.
- Resolved critical SQL and application bugs during UAT and testing phases, reducing defect backlog and accelerating time-to-production for key features.

Stack: Java, Spring Boot, SQL, JavaScript, Linux, Git

Data Science Intern · Deznolitics Web and Software Company

Aug 2022 – Oct 2022 · Navi Mumbai, IN

- Built an end-to-end customer churn prediction pipeline (logistic regression + decision tree ensemble, scikit-learn) on **200,000+ subscription records**; optimised for AUC score and delivered risk scores surfaced via Tableau dashboards adopted by the marketing team for retention targeting.
- Designed and implemented ETL pipelines including data cleaning, feature engineering, and model tuning; Tableau dashboards were used directly in marketing retention strategy decisions.

Stack: Python, scikit-learn, TensorFlow, Tableau, SQL, ETL pipelines

EDUCATION

M.Sc. Computer Science · Universität Bonn

Apr 2024 – Present · Bonn, DE

Current Grade: 1.7 · Focus: Machine Learning, NLP, Large Language Models

B.E. Computer Engineering · Mumbai University

Jun 2018 – Mar 2022 · Navi Mumbai, IN

CGPA: 9.76 / 10 · Final-year thesis on real-time crowd density estimation using CNN/MCNN (~90% accuracy)

PERSONAL PROJECTS

26/27 Football Season Simulator *React · TypeScript · Zustand · Web Workers · Tailwind*

- Built a full browser-based football management simulator ([26-27-football-season-simulator.vercel.app](#)) modeling transfers, squad building, and full 38-match seasons for 12 European clubs (~3,500 real players) using a Dixon-Coles + Monte Carlo match engine and a full domestic/European cup engine, all running client-side on seeded PRNGs.

Got You CoVered - AI Cover Letter Generator *Next.js · TypeScript · Anthropic Claude API · Tailwind · Vercel*

- Built and deployed a **live** full-stack web app ([got-you-covered-cv.vercel.app](#)) that generates tailored one-page cover letters from a user's resume and a target job description using the Anthropic Claude API - supporting custom instructions, an optional reference cover letter to emulate the user's voice, tone/length controls, on-the-spot editing of the generated letter and one-click .docx/.pdf export, with all user data kept client-side.

Value Alignment via Domain-Specific LoRA Fine-Tuning *Python · PyTorch · Qwen3 · LoRA/PEFT · Hugging Face*

- Researched the effect of domain-specific pretraining on value orientations of Qwen3 models (0.6B-8B) using LoRA fine-tuning on five subsets of The Pile; evaluated value shifts against the Kiesel Value Survey (Schwartz theory). Found that shifts in smaller models were driven primarily by output homogenisation rather than substantive value acquisition - a finding with direct implications for responsible LLM deployment.

Multilingual Narrative Classification in News Articles *Python · PyTorch · RoBERTa · LLaMA-3.1-8B · Hugging Face*

- Built a propaganda and narrative detection system for English/Russian news using a dual-head RoBERTa multi-label classifier with LLaMA-3.1-8B-Instruct refinement for low-resource classes. Achieved **85% multi-label accuracy** on a highly imbalanced dataset - approximately 18pp above baseline - enabling automated monitoring of disinformation narratives at scale.

AI-Assisted Agile Planning & Analytics Dashboard *Python · Pandas · Streamlit · REST APIs · Prompt Engineering*

- Designed and prototyped a dashboard to analyse Jira-style sprint data (epics, capacity, dependencies, delivery risks) and generate AI-assisted planning summaries, backlog refinement inputs, and leadership reports - converting raw project data into structured decision support for engineering leads.

Financial Portfolio Analysis Dashboard *Tableau · Python · Yahoo Finance API · Excel*

- Designed a Tableau dashboard comparing portfolio performance across equities, bonds, and ETFs using historical Yahoo Finance data; integrated dynamic filters and comparative charts enabling side-by-side risk/return analysis - demonstrating applied financial analytics and BI skills directly relevant to investment or fintech contexts.

TECHNICAL SKILLS

Languages	Python, Java, JavaScript, C/C++, Shell, SQL, R
ML & Data Science	PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, RoBERTa, Embeddings, Pandas, NumPy
AI Engineering & Gen AI	Generative AI, RAG Pipelines, LLaMA, Qwen3, LoRA (PEFT), Vector DBs (FAISS, Chroma), LangChain, Prompt Engineering, Claude Code, Anthropic Claude API, Google Gemini, OpenAI Codex, AI Agents (CrewAI, AutoGPT, LangGraph), MCP
Data & Viz	Tableau, Power BI, Google Data Studio, Streamlit, Excel
Backend & APIs	Spring Boot, REST APIs, MySQL, MongoDB, Firebase, Docker, Kubernetes (familiar)
Tools, DevOps & Automation	Git, Jira, Linux, Postman, Confluence, LaTeX, Microsoft Office, Zapier, n8n, Make (Workflow Pipelines)
Agile / PM Analytics	Sprint Planning, KPI Tracking, Capacity Analysis, Backlog Refinement, Dependency Tracking

CERTIFICATIONS

Google Data Analytics Professional Certificate (Coursera) · IBM Data Science Professional Certificate (Coursera)

Google Business Intelligence Professional Certificate (Coursera) · Data Visualization with Tableau (UC Davis, Coursera)

ADDITIONAL INFORMATION

Languages: English (Fluent, IELTS 7.5) · Hindi (Fluent) · German (A2, actively improving)

Interests: Football analytics · Chess · Photography · Travelling · Gaming